

RELIEFSPHERE — DATABASE TABLE DESIGN

Complete Collection Schemas & Data Dictionary Specification

STUDENT NAME Amal Babu (INMCA 2022–27)	COURSE CODE 20INMCA509 — Mini Project 2	EVALUATION DATES 18th & 19th August 2026 (RS 405)
DATABASE ENGINE MongoDB NoSQL with Mongoose ODM	TOTAL COLLECTIONS 8 Core Collections	RELATIONAL INTEGRITY Mongoose Schema ObjectIds

1. System Collections Summary

COLLECTION NAME	PRIMARY KEY	FOREIGN KEY REFERENCES	DESCRIPTION
users	_id	—	Central authentication, user credentials, roles, and verification states
donors	_id	userId → users._id	Donor profile, donor classification type, preferred categories, and address
volunteers	_id	userId → users._id	Volunteer skills, ID document verification, and task completion metrics
recipientorganizations	_id	userId → users._id	NGO profile, government registration details, and verification certificates
donations	_id	donorId → donors._id , postedBy → users._id	Donated items, quantity, category, pickup coordinates, and match states
requirements	_id	organizationId → recipientorgs._id	Resource demands posted by recipient organizations with urgency weighting
deliveries	_id	donationId → donations._id , volunteerId → volunteers._id	Logistics assignment, tracking status, and Proof of Delivery (PoD) uploads
notifications	_id	userId → users._id	In-app push notifications and state transition logs for all active users

2. Detailed Field Specifications

2.1 Collection: users					Authentication & System Accounts
Stores user credentials, role definitions, password hashes, and security reset tokens.					
FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION	
_id	ObjectId	PRIMARY KEY	Auto-generated	Unique user identity key	
fullName	String	REQUIRED	—	User's full legal name	
email	String	REQUIRED, Unique	—	Unique email address for login	
password	String	Optional (Google OAuth)	—	Bcrypt hashed password string	
phone	String	Optional	""	Primary contact phone number	
role	String	REQUIRED, Enum	—	donor recipient_org volunteer admin	
isVerified	Boolean	Optional	false	Account verification state	
avatar	String	Optional	""	Profile image URL / path	
resetPasswordToken	String	Optional	null	Password reset token string	
resetPasswordExpires	Date	Optional	null	Token expiry timestamp	
createdAt / updatedAt	Date	Auto-managed	Current Date	Mongoose timestamp tracking	

2.2 Collection: donors					Donor Profiles
Stores extended details for donors including geographic location and preferred categories.					
FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION	
_id	ObjectId	PRIMARY KEY	Auto-generated	Unique donor profile ID	
userId	ObjectId	FK → users_id	—	Refers to base user account	
donorType	String	Enum	"individual"	individual small_business educational_institution	
preferredCategories	Array[String]	Optional	[]	List of preferred item categories	
address	String	REQUIRED	—	Physical street address	
latitude / longitude	Number	Optional	null	GPS coordinates for proximity calculation	
totalDonations	Number	Optional	0	Cumulative completed donation count	

2.3 Collection: volunteers					Volunteer Profiles & Verification
Tracks volunteer skills, government ID uploads, and admin verification status.					
FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION	
_id	ObjectId	PRIMARY KEY	Auto-generated	Unique volunteer profile ID	
userId	ObjectId	FK → users_id	—	Refers to base user account	
dob	Date	REQUIRED	—	Date of birth	
skills	Array[String]	Enum Array	[]	Driving, First Aid, Logistics, Inventory	
idDocument	String	Optional	""	Path to uploaded government ID proof	
verificationStatus	String	Enum	"pending"	pending verified rejected	
approvedBy	ObjectId	FK → users_id	null	Admin ID who approved the profile	
completedDeliveries	Number	Optional	0	Count of successfully completed deliveries	

2.4 Collection: recipientorganizations				NGO & Organization Profiles
Stores beneficiary organization registrations, NGO certificates, and approval status.				
FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
_id	ObjectId	PRIMARY KEY	Auto-generated	Unique organization profile ID
userId	ObjectId	FK → users_id	—	Refers to admin account for org
orgName	String	REQUIRED	—	Official organization name
orgType	String	REQUIRED , Enum	—	ngo orphanage old_age_home government_school
registrationNumber	String	Optional	—	Government registration certificate number
verificationDocs	Array[String]	Optional	[]	Array of file paths to uploaded certificates
verificationStatus	String	Enum	"pending"	pending verified rejected

2.5 Collection: donations				Resource Contributions
Stores goods donated by users, AI match calculations, and lifecycle states.				
FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
_id	ObjectId	PRIMARY KEY	Auto-generated	Unique donation record ID
donorId	ObjectId	FK → donors_id	—	Refers to donor profile
category	String	REQUIRED , Enum	—	food clothes books medicine essentials
donationName	String	REQUIRED	—	Item title / description
quantity	Number	REQUIRED , Min: 1	—	Quantity of items offered
status	String	Enum	"pending"	pending matched accepted assigned picked_up delivered
matchScore	Number	Optional	null	Calculated 100-point AI match score

2.6 Collection: requirements

Beneficiary Demand Requests

Stores resource needs created by verified organizations with EDF urgency levels.

FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
<code>_id</code>	ObjectId	PRIMARY KEY	Auto-generated	Unique requirement ID
<code>organizationId</code>	ObjectId	FK → <code>recipientorgs._id</code>	—	Refers to recipient organization
<code>category</code>	String	REQUIRED, Enum	—	food clothes books medicine essentials
<code>quantity</code>	Number	REQUIRED, Min: 1	—	Quantity needed
<code>urgency</code>	String	Enum	"medium"	low medium high critical
<code>status</code>	String	Enum	"pending"	pending open matched fulfilled closed

2.7 Collection: deliveries

Logistics & Volunteer Operations

Tracks dispatch tasks, volunteer assignment, and Proof of Delivery photo verification.

FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
<code>_id</code>	ObjectId	PRIMARY KEY	Auto-generated	Unique delivery record ID
<code>donationId</code>	ObjectId	FK → <code>donations._id</code>	—	Refers to matched donation
<code>volunteerId</code>	ObjectId	FK → <code>volunteers._id</code>	null	Refers to assigned volunteer
<code>status</code>	String	Enum	"assigned"	assigned picked_up in_transit delivered failed
<code>proofImages</code>	Array[String]	Optional	[]	Paths to uploaded Proof of Delivery photos
<code>proofNote</code>	String	Optional	""	Delivery notes / comments by volunteer

2.8 Collection: notifications

System Alert Logs

Stores user notifications for system state changes and recommendation matches.

FIELD NAME	DATA TYPE	CONSTRAINTS	DEFAULT	DESCRIPTION
_id	ObjectId	PRIMARY KEY	Auto-generated	Unique notification ID
userId	ObjectId	FK → users._id	—	Target user ID
title	String	REQUIRED	—	Notification header title
message	String	REQUIRED	—	Detailed notification message text
type	String	Enum	"system"	donation requirement delivery system match
isRead	Boolean	Optional	false	Read/unread status flag